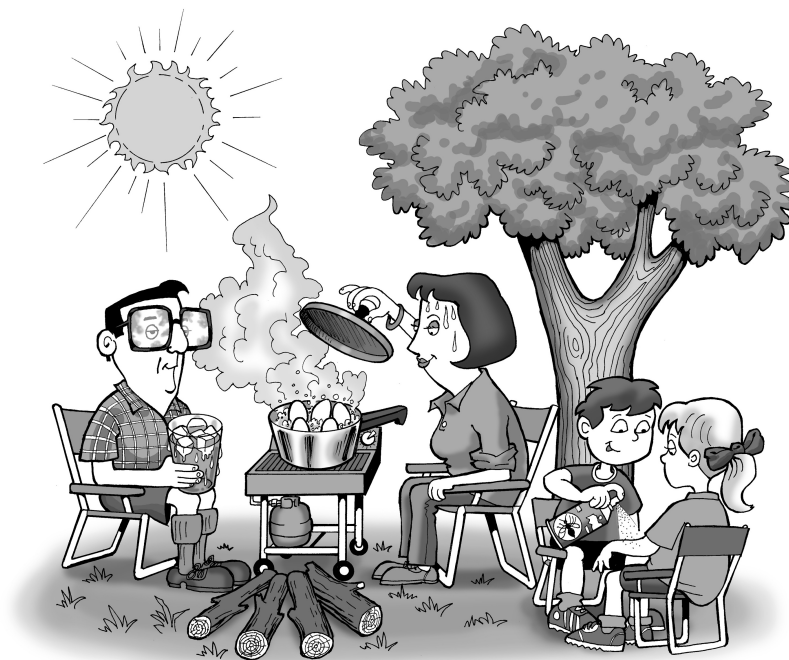


Campsite chillout

Answers

Even while relaxing on holiday, you can find exothermic and endothermic processes going on all around you.

- For each of the activities shown on the illustration and listed below, state whether the process indicated is exothermic or endothermic, and then explain your choices. (Remember: an exothermic process results in thermal energy being given out to the surroundings, while an endothermic process requires thermal energy which is drawn in from the surroundings.)



- The leaves use energy from the sun to make sugars for the tree (photosynthesis).
Endothermic — energy is being absorbed.
 - Nuclear reactions in the sun warm the Earth during the day.
Exothermic — energy is being released.
 - Sweat on a person's face evaporates.
Endothermic — energy is being absorbed. Change of state: liquid → gas.
 - The gas barbecue burns.
Exothermic — energy is being released
 - The eggs inside the saucepan cook to become hard-boiled.
Endothermic — energy is being absorbed
 - Glasses fog up from the boiling water.
Exothermic — energy is being released. Change of state: gas → liquid.
 - Insect repellent is liquid inside the can, partly gaseous as it comes out of the nozzle.
Endothermic — energy is being absorbed. Change of state: liquid → gas.
 - Ice melts in a drink.
Endothermic — energy is being absorbed. Change of state: solid → liquid.
- Which of these processes can be classified as chemical reactions? Explain your choices fully.
Chemical reactions are irreversible and involve the formation of a new substance.
 - Photosynthesis: carbon dioxide + water → glucose + oxygen
 - This is an example of a nuclear reaction in which the internal structure of the atom changes and different types of atoms are formed.
 - Combustion of methane: methane + oxygen → carbon dioxide + water
 - The proteins in the egg change (denature and coagulate). This is irreversible.